TrueDeed

Algorithmic Land Fraud & Double-Selling Prevention Platform

PROBLEM P3: CATCH FRAUD & DOUBLE-SELLING

LandTech 2026 Innovation Hackathon

Team Code: LA-72 (Team TrueDeed)	**Track:** Digital Land Records, Verification & Digitization	**Document:** Concept Note (Max 4 Pages)

1. PROBLEM CHOSEN: P3 — CATCH FRAUD & DOUBLE-SELLING

We have selected **Problem P3: Catch Fraud & Double-Selling** as defined in Section 5 of the official LandTech 2026 Rule Book: *"Spot fake documents, the same land sold to more than one buyer, and records that do not match between registration and mutation."*

2. WHY IT MATTERS IN BANGLADESH (NATIONAL URGENCY)

Land disputes are the single greatest driver of social instability, familial violence, and judicial congestion in Bangladesh. Over **70% of the 3.2+ million civil litigations** currently pending in national courts originate from fraudulent land titles, trapping generations of citizens in perpetual legal warfare and immobilizing billions of takas in productive capital.

> **The Core Institutional Flaw:** Deed Registration is governed by the *Directorate of Registration under the Ministry of Law*, whereas Record-of-Rights (Khatian surveys and Mutation) are administered by Upazila AC Land offices under the *Ministry of Land*. These two jurisdictions operate without real-time synchronization.

Fraud syndicates exploit this structural disconnect through three disastrous mechanisms:

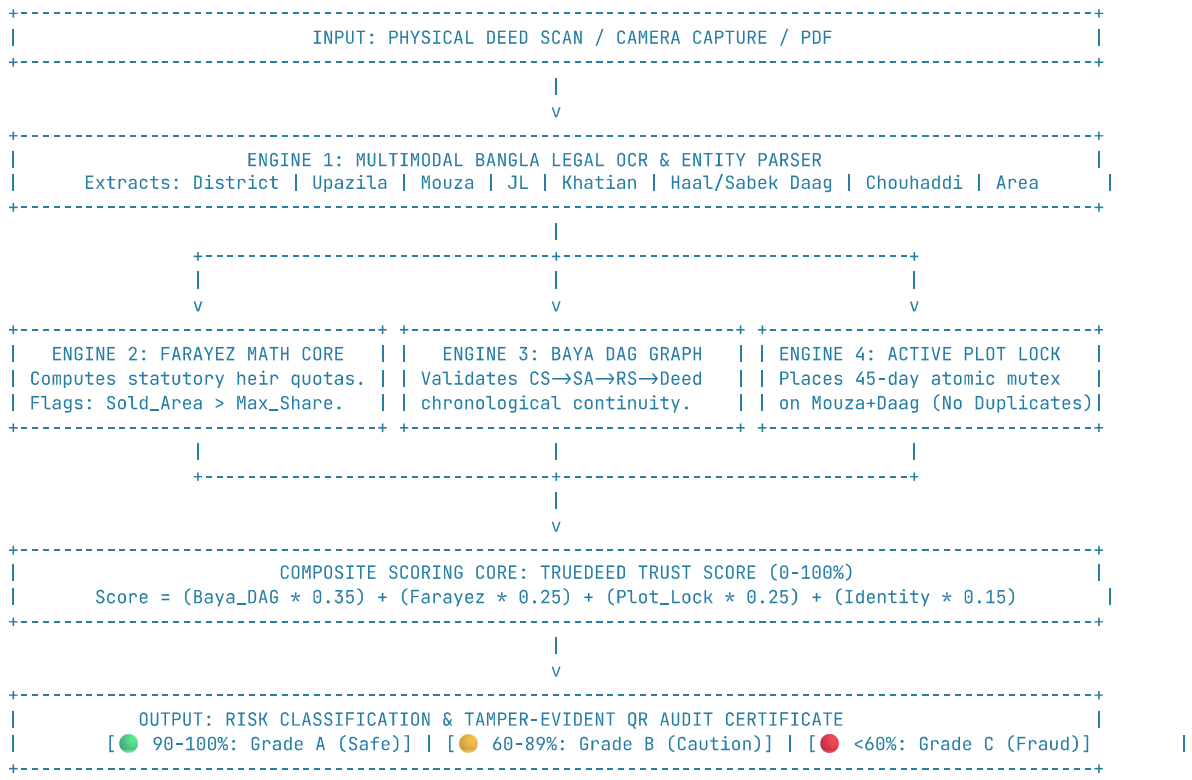
- **Inheritance Over-Selling (Farayez Violation):** Following an ancestor's death, heirs inherit strict statutory quotas under the *Muslim Family Laws Ordinance 1961*. Fraudulent heirs routinely conceal female siblings, fabricate unauthorized local heir certificates, and sell 30 to 40 decimals from an estate where their mathematical share is only 15 decimals.
- **Broken Baya Lineage (Fake Preceding Deeds):** Transfer deeds must establish an unbroken chain of ownership connecting back to state survey records (CS → SA → RS) under Section 17A of the *Registration Act 1908*. Fraudsters inject fabricated intermediate deeds featuring fictitious sub-registry volume books that cannot be verified manually by common buyers.
- **The 45-Day Double-Selling Window:** Under Section 143 of the *State Acquisition and Tenancy Act 1950*, finalizing an e-Mutation (নামজারি) requires 30 to 45 days post-registration. In this latency window, corrupt sellers execute duplicate sales of the identical plot to multiple buyers before the first sale reflects on government servers.

3. PROPOSED SOLUTION: THE TRUEDEED INFRASTRUCTURE

TrueDeed is an automated algorithmic verification layer designed to bridge the void between Sub-Registry offices and AC Land offices. Rather than acting as a passive record query portal like *e-Porcha*, TrueDeed provides **pre-execution fraud intercept algorithms** that mathematically evaluate title integrity, statutory inheritance quotas, and temporal transaction locks in under 5 seconds before deeds are executed.

4. HOW IT WORKS: SYSTEM ARCHITECTURE & WORKFLOW

TrueDeed transforms physical deed scans into mathematical proofs of title authenticity via a 4-engine microservice architecture:



5. CORE TECHNOLOGY STACK & FEASIBILITY

TrueDeed is architected for immediate zero-cost execution and maximum deployment feasibility:

- **Backend Framework:** Python (FastAPI) delivering asynchronous, sub-second REST microservices.
- **Perception & Vision AI:** OpenCV for adaptive binarization, noise filtering, and deskewing, paired with Google Gemini 2.0 Flash Vision API for multimodal Bengali legal entity extraction.
- **Frontend Interface:** Next.js 14, React, and Tailwind CSS engineered for an accessible, bilingual (Bangla/English) citizen and bank underwriter dashboard.
- **Data & Concurrency Layer:** SQLite for embedded prototyping and PostgreSQL with PostGIS schemas for spatial land coordinates and temporal mutex locks.
- **Cryptographic Security:** SHA-256 hashing to generate immutable digital verification audit certificates.

> **Zero-Cost & Privacy Compliance:** In accordance with Section 10 of the LandTech Rule Book, TrueDeed operates entirely on open-source libraries and enterprise free tiers. All testing and demonstrations utilize **strictly synthetic dummy records** modeled after official Sub-Registry deed templates.

6. MAIN FEATURES OF TRUEDEED

Feature 1: Bangla Legal Document OCR & Entity Extraction

Eliminates manual form entry by scanning physical deed documents (handwritten or printed) and directly extracting key structural metadata: District, Upazila, Mouza, JL Number, CS/SA/RS Khatian numbers, Haal and Sabek Daag numbers, directional four boundaries (চৌহদ্দি), and transferred decimal area.

Feature 2: Algorithmic Farayez (Inheritance Math) Validator

Implements statutory mathematical rules of the *Muslim Family Laws Ordinance 1961* ($1 \text{ Son} = 2 \text{ Daughters}$). Given the ancestor's baseline land volume and surviving heirs, it calculates each heir's exact mathematical ceiling. If `Transferred_Area > Eligible_Share`, the engine triggers an instant **Critical Over-Selling Alert**.

Feature 3: Directed Acyclic Graph (DAG) for Historical Baya Lineage

Models historical title deeds as a Directed Acyclic Graph connecting original survey records (CS → SA → RS → Mutation → Current Deed). Automatically validates: (1) *Temporal Consistency*, (2) *Party Continuity* (Buyer in Deed N matches Seller in Deed $N+1$), and (3) *Area Conservation*.

Feature 4: 45-Day Temporal Active Plot Lock (Anti-Double Selling)

Enforces a distributed temporal mutex lock on the composite geographic identifier (`District_Upazila_Mouza_Khatian_Daag`) upon transaction initiation. Any secondary transaction attempted on the same plot within the 45-day mutation processing window triggers an immediate **Concurrent Transfer Warning**.

Feature 5: Objective TrueDeed Trust Score (0–100%)

Aggregates sub-engine validations into a single standardized risk metric: Baya Chain Continuity (35%), Farayez Share Compliance (25%), Active Plot Lock (25%), and Identity/Record Consistency (15%). Active Section 144 injunctions trigger a hard fail-safe override to < 30%.

7. THE TRUEDEED TRUST SCORE MATRIX

EVALUATION METRIC	WEIGHTAGE	ALGORITHMIC VERIFICATION STANDARD
Baya Chain Integrity	**35 Points**	Unbroken DAG continuity from state baseline (CS/SA/RS) to current deed.
Farayez Share Accuracy	**25 Points**	Transferred acreage is strictly ≤ statutory inheritance entitlement.
Active Plot Lock	**25 Points**	Zero concurrent transfer filings on target Daag within the 45-day window.
Registry & NID Match	**15 Points**	Cross-verified consistency across Seller NID, Khatian, and Sub-Registry logs.
Hard Veto Override	**FAIL-SAFE**	Active Section 144 court injunction triggers an immediate drop to < 30%.

8. NATIONWIDE DEPLOYMENT & E-GOVERNANCE ADOPTION PLAN

TrueDeed is designed to integrate seamlessly into Bangladesh's national e-governance framework:

STAKEHOLDER SECTOR	NATIONWIDE E-GOVERNANCE DEPLOYMENT VALUE
Sub-Registry Offices	Integrated directly into the deed registration software at Sub-Registry desks to verify ownership and statutory heir quotas before printing stamp papers.
AC Land Offices	Automates cross-checking during e-Mutation hearings, instantly catching forged Baya deeds and preventing duplicate mutation applications.
Mortgage Banks & NBFIs	Integrated via secure B2B API into loan origination systems to reduce collateral property title vetting from 21 days down to 5 seconds.
Citizens & Expatriates	Mobile-first citizen portal ("Check Before You Pay") enabling non-resident Bangladeshis and buyers to verify title validity before paying token money.

9. TEAM COMPOSITION & MENTORS (RULE BOOK COMPLIANCE)

In strict compliance with Section 6 of the LandTech 2026 Rule Book, Team TrueDeed comprises 2 DIU students and 2 specialized mentors:

ROLE	NAME	AFFILIATION / ID	RESPONSIBILITY
Student 1 (Team Leader)	Ishtiak Ahmed	DIU Student	System Architecture, Full-Stack Engine & Presentation
Student 2 (Core Member)	[Teammate Name]	DIU Student	UI/UX Design, Concept Documentation & Testing
Faculty Mentor	[Faculty Mentor Name]	Department of CSE, DIU	Academic Guidance, Algorithmic Validation
Industry Mentor	[Industry Mentor Name]	Industry Expert / Tech Lead	GovTech Feasibility & Enterprise Scaling

10. HACKATHON EXECUTION ROADMAP

- **Round 1 (Current Submission):** Concept Note, Algorithmic Specifications, System Architecture, and GitHub repository.
- **Round 2 (Final On-Site):** Live interactive prototype demonstrating real-time deed upload, Farayez fraud detection, Baya DAG timeline, and Trust Score generation.
- **National Vision:** Pilot collaboration with Ministry of Land and ICT Division for national rollout.